

2.2 Linear Regression

Introduction

Linear Regression is one of the most fundamental and widely used algorithms in Supervised Machine Learning. It is mainly used for regression problems, where the goal is to predict a continuous numerical value based on one or more input features. Linear regression establishes a linear relationship between independent variables (features) and a dependent variable (output).

Definition of Linear Regression

Linear Regression is a statistical and machine learning technique that models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data.

The general idea is:

$$\text{Output} = (\text{Weight} \times \text{Input}) + \text{Bias}$$

2.2.1 Simple Linear Regression

Introduction

Simple Linear Regression is a type of linear regression where there is only one independent variable used to predict a dependent variable.

Mathematical Representation

The equation of Simple Linear Regression is:

$$y = mx + c$$

OR

$$y = \beta_0 + \beta_1 x$$

Where:

- y = Dependent variable (output)
- x = Independent variable (input feature)
- β_0 / c = Intercept (constant term)
- β_1 / m = Slope of the line

Slope (m)	Relationship Type
$m > 0$	Positive
$m < 0$	Negative
$m = 0$	No relationship

Working of Simple Linear Regression

1. Collect dataset with one input feature and output
2. Plot data points on a graph
3. Fit a straight line that best represents the data
4. Minimize error using a loss function (Mean Squared Error)
5. Use the line to make predictions

Example

Predicting **student marks** based on **number of hours studied**.

Hours Studied (x)	Marks (y)
2	40
3	45
4	55
6	70

Advantages of Simple Linear Regression

- Easy to understand and interpret
- Computationally efficient
- Useful for basic prediction problems

Limitations of Simple Linear Regression

- Assumes linear relationship
- Sensitive to outliers
- Not suitable for complex data

Problem: Predict Student Marks based on Study Hours

Problem Statement - to predict **exam marks** based on **number of study hours** using Linear Regression.

Sample Dataset-

Study Hours	Marks
1	35
2	40
3	50
4	55
5	65
6	70
7	75
8	85

✓ This is Supervised Learning

Because we have labeled data (Hours → Marks)

✓ **It is Regression** - Output (Marks) is continuous numeric value **i.e** output is numeric

✓ **Model Equation:** $y = mx + c$

$$\text{Marks} = m \times \text{Hours} + c$$

- m = slope
- c = intercept

✓ **Goal:** Minimize error between actual and predicted marks

Python Code -

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```
# Step 1: Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Step 2: Create dataset
hours = np.array([1,2,3,4,5,6,7,8]).reshape(-1,1)
marks = np.array([35,40,50,55,65,70,75,85])

# Step 3: Create model
model = LinearRegression()

# Step 4: Train model
model.fit(hours, marks)

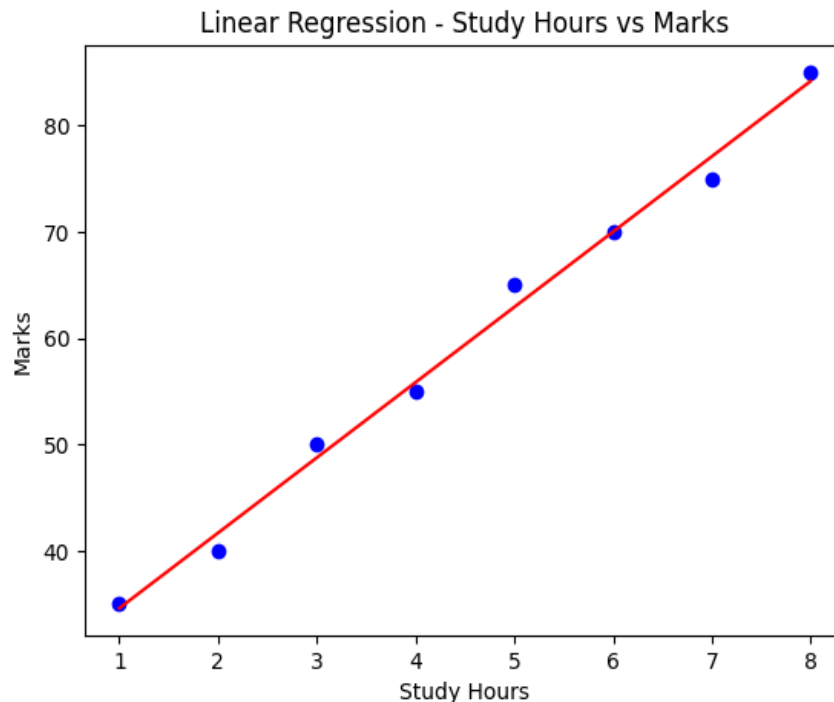
# Step 5: Get slope and intercept
print("Slope (m):", model.coef_)
print("Intercept (c):", model.intercept_)

# Step 6: Predict marks for 9 hours
predicted = model.predict([[9]])
print("Predicted Marks for 9 hours:", predicted)

# Step 7: Plot graph
plt.scatter(hours, marks, color='blue')
plt.plot(hours, model.predict(hours), color='red')
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Linear Regression - Study Hours vs Marks")
plt.show()
```

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```
Slope (m): [7.08333333]
Intercept (c): 27.499999999999996
Predicted Marks for 9 hours: [91.25]
```



- Blue dots → Actual data points
- Red line → Best fit regression line
- Line shows positive linear relationship
- As study hours increase → marks increase.

Output Explanation

slope (m) = 7.08

For every 1 hour increase in study time, Marks increase by approximately 7.14 marks.

Intercept (c) = 27.5

This means: If a student studies 0 hours, Expected marks ≈ 27.5

Prediction for 9 Hours

If student studies 9 hours:

Marks = $7.14 \times 9 + 27.5$ = So model predicts around **91 marks**.